

KevaBook Domain Model

1. Domain Overview

The KevaBook system is designed around a multi-tenant domain model in which each **Business** represents an isolated tenant. Each Business manages its own staff, services, availability schedules, bookings, locations, and configuration settings independently of other businesses.

The domain model provides a logical representation of the core business concepts and their relationships. It is independent of implementation details such as database design, microservice architecture, or technology choices.

2. Core Domain

The KevaBook domain is centered around the **Business** entity, which represents an independent service provider operating on the platform.

The Business domain encompasses the following core entities:

- **Staff** – Individuals associated with a Business who are responsible for delivering services.
- **Services** – Offerings provided by a Business that can be booked by clients.
- **Availability** – Time-based schedules defining when Staff members are available for bookings.
- **Bookings** – Client-generated reservations linked to Services and assigned to specific Staff members.
- **Location** – Physical or virtual service locations associated with a Business.
- **Settings** – Configuration parameters specific to each Business tenant.

3. Core Domain Entities

3.1 Business

The Business entity represents a company registered on the KevaBook platform and serves as the tenant boundary for all associated data.

Key Fields:

- businessId
- name
- description
- email
- phone
- status
- createdAt
- updatedAt

Relationships:

- One Business has many Staff members.
- One Business offers many Services.
- One Business may have multiple Locations.
- One Business has many Bookings.
- One Business is managed by one or more Business Owners.

3.2 Staff

The Staff entity represents an individual associated with a Business who is responsible for delivering services.

Key Fields:

- staffId
- businessId
- fullName
- email
- phoneNumber
- role
- active
- createdAt
- updatedAt

Relationships:

- Each Staff member belongs to one Business.
- One Staff member has many Availability schedules.
- One Staff member may have many Bookings.

3.3 Service Catalog

The Service entity defines a bookable offering provided by a Business.

Key Fields:

- serviceId
- businessId
- name
- description
- durationMinutes
- price
- currency
- active

Relationships:

- Each Service belongs to one Business.
- One Service may be associated with many Bookings

3.4 Availability (Staff Schedule)

The Availability entity defines the working schedule of a Staff member.

Key Fields:

- availabilityId
- staffId
- dayOfWeek
- startTime
- endTime

Relationships:

- Each Availability record belongs to one Staff member.

3.5 Booking

The Booking entity is the central transactional object in the system. It represents a confirmed or pending reservation of a Service provided by a Staff member within a Business.

Key Fields:

- bookingId
- businessId
- staffId
- serviceId
- clientName
- clientEmail
- clientPhone
- startTime
- endTime
- status
- createdAt
- updatedAt

Relationships:

- Each Booking belongs to one Business.
- Each Booking is assigned to one Staff member.
- Each Booking is associated with one Service

3.6 Client

The Client entity represents a customer who books services through the platform. Clients may book appointments without creating an authenticated account.

Key Fields:

- clientId
- name
- email
- Phone

Relationships:

- One Client may have many Bookings

3.7 Location

The Location entity represents a physical or virtual location where services are delivered.

Key Fields:

- locationId
- businessId
- street
- city
- country

Relationships:

- Each Location belongs to one Business.

4. Domain Relationships

The relationships between the core domain entities are summarized below:

- A Business manages multiple Staff members.
- A Business offers multiple Services.
- A Business may operate from multiple Locations.
- Staff members define their Availability schedules.
- Clients create Bookings for Services.
- Each Booking is assigned to a single Staff member.
- Each Booking belongs to one Business.
- Business Owners manage the Business and its associated entities.